

9 (**short**) Some uncracked bicycle forks are subject to fatigue loading. Approximate $S - N$ data for the material used are given in Fig. 9, for zero mean stress.

(a) The loading cycle due to road roughness is assumed to have a constant stress range $\Delta\sigma$ of 1200 MPa and a mean stress σ_m of zero. How many loading cycles will the forks withstand before failing? [2]

(b) Due to a constant rider load the mean stress is 100 MPa. Use Goodman's rule to estimate the percentage reduction in lifetime associated with this mean stress. The tensile strength σ_{ts} of the steel is 1100 MPa. [4]

(c) Define the fatigue limit. What practical changes could be made to the forks to bring the stress range below the fatigue limit and avoid fatigue failure? [2]

(d) Closer inspection of a failed fork reveals that the failures occur where the forks increase in section. Explain why failure occurs at this location, and suggest an alternative change in fork design to avoid failure. [2]

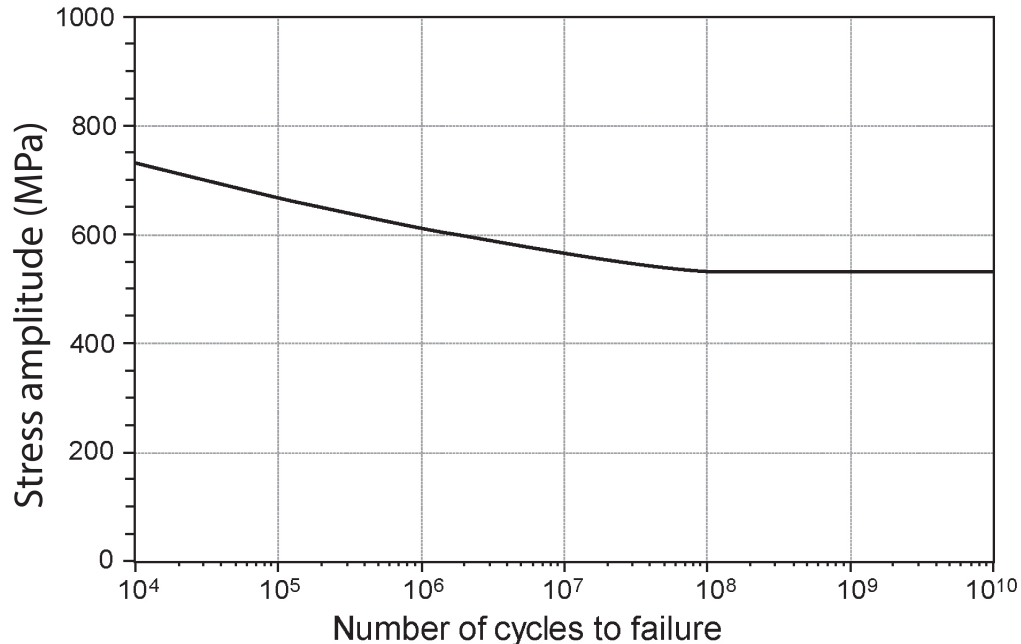


Fig. 9